



Addis Abeba University

School of Health Science

ORGANIC CHEMISTRY LABORATORY GROUP REPORT

<u>GROUP MEMBERS</u>	<u>ID NUMBERS</u>
1.SAMKET FASIL.....	UGR/3649/13
2.ROZA SEIFU	UGR/ 0781/13
3.TSEGANESH AYALEW	UGR/7606/13
4.RAHMET ARAGIE	UGR/6117/13
5.YEABSIRA ENDALE	UGR/ 7522 /13
6. NATHNAEL DEMISSIE	UGR/6617/13
7.EYOB YONAS.....	UGR/2078/13
8. EYOB ABEBE.....	UGR/8276/13
9.	

submission date :-Feb,8,2022

Experiment number: 1

Title: Recrystallization

Objective:- This experiment is conducted in order to purify a contaminated solid compound which is benzoic acid from its impurities by the technique of recrystallization.

Theory

- ❖ The purification of solids by crystallization is based upon differences between solubility of desired solid and its impurity in a given solvent which have consecutive crystallization processes including:
 - Dissolving the impure substance with suitable solvent at or near the boiling point
 - allowing the hot solution to cool causing the dissolved substance to crystallize out
 - separate the crystals from the solution
- ❖ At this experiment the solution was filtered while it was hot to remove insoluble impurities and water was used as a solvent while benzoic acid as a solute.

Apparatus used

- 4g of benzoic acid(impure solute)
- 20ml of water(solvent)
- Erlenmeyer flask
- Measuring cylinder
- Steam less funnel
- Fluted filter paper
- Suction filter(formed by assembling Buchner funnel+ suction flask)
- Hot plate
- Water aspirator
- Paper box

PROCEDURE

- First 4g of impure Benzoic acid was already measured and prepared.
- Then the 4g of impure benzoic acid was added to an Erlenmeyer flask.
- After that 20ml of water was measured and then added to the flask containing the impure benzoic acid.
- The flask containing the impure benzoic acid and water was heated by being placed on a hot plate while the solution was being stirred.
- The impurity was dissolved completely Then the solution was poured slowly into a flask with a stemless funnel on it containing a filter paper and only the impurity was left .
- The flask was put inside ice to let it cool and crystalize.After a few minutes, the flask was taken out of the ice.
- A new fluted filter was weighted which becomes **0.59g**,Then the folded fluted paper was placed on a Buchner funnel.
- The Buchner funnel was attached to the suction filter, a water aspirator was connected to it, and the tap water was turned on.
- Then the crystalized solution was added onto the Buchner funnel containing the fluted filter paper
- paper box was prepared,the filter paper containing the filtered sample was placed on it.

RESULTS

During our experiment in conductig crystallization of benzoic acid the weights of the used apparatus and amount of used chemical were recorded to see the final results.

- The weight of impure benzoic acid= 4g
- volume of water=20ml
- The weight of filtered paper= 0.59g
- The weight of dried filter paper +sample=4.3g

Therefore :- weight of the purified solute = $4.3\text{g} - 0.59\text{g} = 3.71\text{g}$

% impurity=(mass of pure benzoic acid÷ initial mass of crude/ impure crystal) × 100

$$= (3.71\text{g} \div 13\text{gm}) \times 100 = 28.5\%$$

CONCLUSION:- .

From the process of recrystallization, one can conclude that impurities which have different solubility rates can be easily distinguished and the solute can also be again recrystallized using the above procedures. The final result reveals that the amount of impurity in a given solute can be known through these steps.

Questions

1. What is the significance of the term “like dissolves like”?

- The principle “like dissolves like” is that substances which have similar intermolecular forces and tend to dissolve completely. For example, a polar solvent tends to dissolve polar solutes and a non-polar solvent tends to dissolve non-polar solutes.

2. What are the differences between an ordinary paper and filter paper?

- Filter paper is more porous and will catch the containment more efficiently than ordinary paper.
 - It increases the speed of filtration.
 - It has a larger surface area through which the solvent can seep.
 - The folds also create space between the filter paper and glass funnel, allowing for displaced air to more easily exit the flask as liquid drains.
- ❖ Normal paper lacks the above qualities, and cannot separate complex material but filter paper can do this.

Experiment 2 -Simple distillation

Introduction

Simple distillation is a process of a liquid mixture to form vapor and then cooling that vapor to get a liquid. It is a method of separating mixtures based difference in their volatilities in boiling liquid mixtures. It works when differences between 2 boiling points are less than 25 degree Celsius. Unless it does not get separated and purified completely.

Objective: To purify a compound by separating it from a nonvolatile or less volatile material.

MATERIALS USED

- Measuring cylinder
- Thermometer
- Distillation flask
- Condenser
- Adapter
- Water bath
- Bunsen burner
- Boiling chips
- Ethyl alcohol (25 ml)
- Funnel

Procedures:

1. 25 ml of ethyl alcohol contaminated with dye and higher boiling point liquid was poured. The distilling flask was filled with it and used extension of clamp on the distilling flask.

2. A few boiling stone or chips were added to the flask.

3. The thermometer bulb was positioned below the arm of 3-way of adapter, where the vapor's turn towards the condenser.

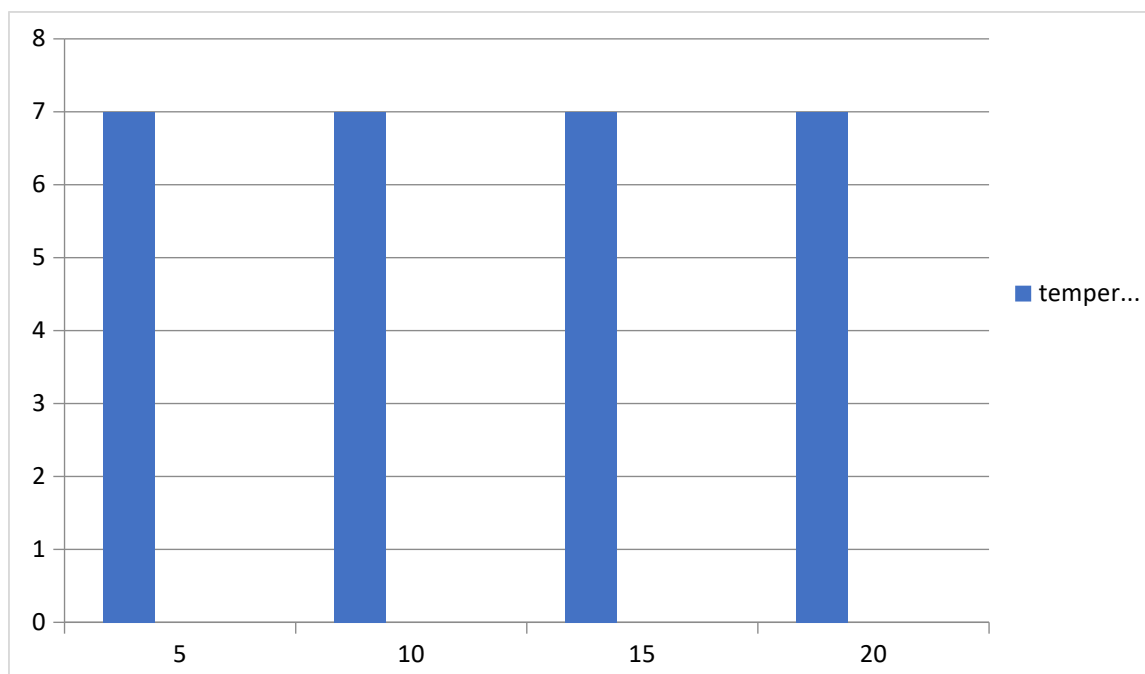
4. Wet condenser hoses with water before being attached.

5. The condenser hoses were connected then the water flow uphill, brought water from faucet into lower arm, and drain out the upper arm.

6. The condenser water turned on.

7. The heat source was applied to the distilling flask by placing it on water bath and heated by means of Bunsen burner.

8. The boiling point that was shown with in every 5 ml was recorded.



This is the volume versus temperature which was recorded when the experiment is done.

Result

This experiment was aimed to purify ethyl alcohol which was first an impure compound. Our evidence for the existence of impurity is red color which has been seen on the beginning of our experiment. For 20 ml with each 5 ml difference the temperature was recorded and it was around 71 degree Celsius with each volume interval. Simultaneously filtered ethyl alcohol was collected on the other flask after it passed through condenser which contains water for the purpose of cooling the vapor and condensing them into liquid droplets. The collected ethyl was become colorless. Finally after we saw these results we conclude that the impurity is filtered due to the color change from red brown to colorless. The other result which has been shown is the temperature remains constant at each interval of volume.

Generally, our experiment show how can we separate the mixture of two miscible with a single cycle of vaporization and condensation.

Quiz question on this experiment

1. Why does not all the liquid vaporize at once when the boiling points reached?

- Because not all molecules go from the liquid to gaseous state immediately. The vapor will form nearest the flame because the heat energy will speed up the kinetic energy of those molecules get closer to the flame. As a side, you may also see bubbles forming on the sides and other parts of the flask. Remember, liquid molecules that are just ready to vaporize are still moving and begin to show in other places then just over the flame. Also, heat will conduct through the glass flask and cause other molecules to vaporize in other places.

2. An organic liquid begins to decompose at 80 degree Celsius. Its vapor pressure at that temperature is at 36 mm. How would you distill the liquid?

- The organic liquid could be distilled by steam distillation, because this distillation is used for high boiling point substances that decompose before the boiling point reached. In this case, instead of using a vacuum, the liquid in questions is mixed with another, immiscible liquid. The presence of the second liquid causes both to boil at a temperature lower than the regular boiling points of either liquid.

3. Why is cold water circulated through a condenser from the bottom upward rather than from top downward?

- Water is circulated from the bottom upward in order to fill the entire condenser. If water enters from the top it drains from bottom and never fills it up. A full condenser is necessary to maximize the heat transfer for condensation. If the water enters through the top and the flow rate is insufficient, the water will merely trickle down one side of the condenser. This would mean only a small section of the condenser would be adequately cooled enough to provide condensing action. By allowing water to enter from the lower hole, the entire volume of the chamber would have to be filled before the water left, provide far more efficient cooling.

4. If the liquid to be distilled had a boiling point higher than 90 degree Celsius, what kind of bath would you use?

- Water baths, heated on a hotplate, are most commonly used to heat solutions. They may also be used to heat to lower temperatures, although it can be difficult to maintain a constant temperature.
- Oil baths are much like water baths, but use silicone or mineral oils in order to enable temperatures hotter than the boiling point of water. so, it is preferable to use an oil bath since the liquid has a high boiling point.

Experiment 3

Title:-survey on Functional Test

Introduction

Functional group are the aggregation of some element that determine the physical and chemical properties of certain compound. One of the physical property we saw under experiment -3 is solubility of the substance. Solubility is the property that depend on polarity and density of substance.

Objective

The core of this experiment is to identify and study the main physical property (solubility)of some functional group.

Theory

Functional groups are a unique combination of atoms present in a molecule that is particularly responsible for the character set of reaction that the molecule exhibit. Some of well know functional group are:alkene, alkane, alkyne,aromatic, carboxylic acid,ketone, aldehyde, ether and ethers. The functional group we saw in Our experiment are Alkane, alkene and , alcohol. Alkane are a compound that have carbon -carbon single bond. These compound is non polar and inactive because of the stability of single bond. Alkene are compound that contain carbon -carbon double bond. It is slightly active but comparing with rest function group it is still inactive. The third functional group we saw under our experiment is alcohol.It is functional group that have hydroxy attachment on it. It has hydrogen bond due to these attachments.

Apparatus used in the experiment

Test tube
Dropper
Stirrer
Test tube holder
Measuring cylinder
Delivery tube
Boss head

Chemical used in experiment

Solute

For Alkane we used N-hexane .

For alkene we used cyclohexene.

For alcohol we used phenol.

Solvent

For Alkane we used: ethanol, water, petroleum ether and concentrated sulphuric acid

For Alkene we used: water, bromine in carbontetrachloride ,potassium permanganate and concentrated sulphuric acid.

For Alcohol we used ferric chloride.

Procedure

All the material and chemical use for the reaction were prepared by the our lecture and assistant. First we wore our laboratory close and glass,then we collect the material and Chemical use for

experiment and then we done the test by the using the following step mentioned by our manual book.

A. Alkane solubility Test

Step. 1 Measure the solute using measuring cylinder. Take 5drops of N-hexane and put in three test tube.

Step. 2 Measure the solvent by taking 3mL of each solvent and add them in test tube. For the solubility by sulphuric acid these steps will change. First we add acid then we add solute.

Step. 3 Shake it well with our with out Stirrer.

B. Alkene Solubility Test

Step.1 Take 0.5mL (10 drops)of cyclohexene and put it in test tube.

Step. 2 Add 3 drops of sulphuric acid, water and bromine in carbontetrachloride and 1mL.

Step. 3 Mix it well.

C. Alcohol Solubility Test

Step. 1 Take 20 drops 10% aqueous solution of phenol and put it in test tube.

Step.2 Add 3 drops of 2% of ferric chloride solution to test tube.

Setp. 3 Mix it well

Result observed

In the Alkane test the N-hexane was solved only in the petroleum ether. In the rest solution there was layer between them and there was no any color change occur. In the Alkene the cyclohexene was dissolved in sulphuric acid and bromine in tetrachloride solution, but there were no Solubility observed with mixing of water and potassium permanganate with the given solute. In the Alcohol test phenol solved with ferric Chloride and gave violet color.

Alkane	water	ethanol	petroleum ether	conc.sulphuric acid
N-hexane	insoluble	insoluble	soluble	insoluble

Alkene	water	conc.sulphuric acid	Bromine in CCl ₄	Potassium permanganate
cyclohexene	insoluble	soluble dark color	solubl Colourless	insoluble

Alcohol	ferric chloride
phenol	soluble Violet color

Conclusion

According to "like dissolve like " polar molecules are dissolved in polar where as the nonpolar dissolved in non polar. The polarity of water, ethanol and sulphuric brings insolubility in non polar solute. Also the layer formulating occur in the solution which were unmixed were due to the difference in density

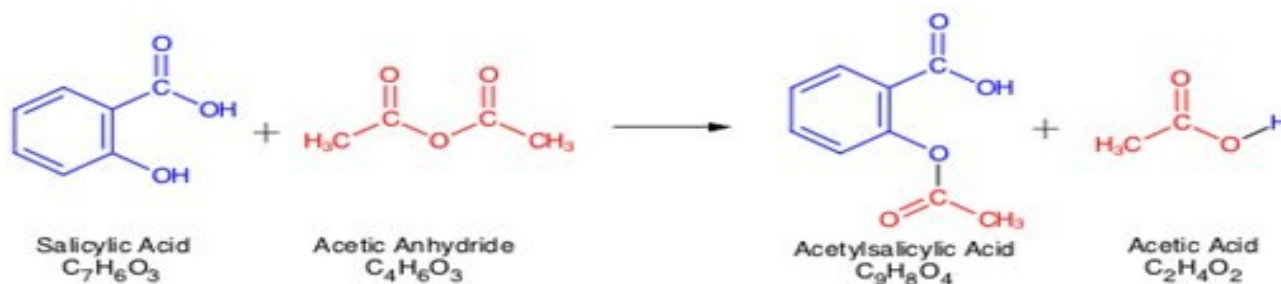
Experiment : 4

Title:-Preparation of Aspirin (Acetylsalicylic acid)

INTRODUCTION

Acetyl salicylic acid(ASA) or Aspirin is widely used as a fever reducer and as a pain killer. To prepare aspirin, salicylic acid is reacted with an excess of acetic anhydride. A small amount of a strong acid is used as a catalyst which speeds up the reaction. In this experiment, sulfuric acid will be used as the catalyst. Warming the reaction mixture is necessary to accelerate the rate of reaction.

The crude product can be isolated and recrystallized to give aspirin as a white crystalline solid.



APPARATUS AND CHEMICALS

- Measuring cylinder
- Test tube
- Ice bath
- Erlenmeyer flask
- Suction or filter flask
- Buchner funnel
- Buckner flask
- Spatula
- Filter paper
- Paper box
- Water
- Salicylic acid
- Acetic anhydride
- Sulfuric acid
- Ethanol
- 10% sodium bicarbonate

PROCEDURE

First 2g of salicylic acid(SA) was placed into a clean dry test tube and 4ml of acetic anhydride was added to it followed by 5 drop of concentrated sulfuric acid.

The mixtures were being swirled gently.

The test tube that contain the mixture of salicylic acid, acetic anhydride and sulfuric acid was being warmed by immersing it partially in a hot water bath for about 10 minutes.

Then the mixture was poured to an Erlenmeyer flask that contain 50ml of water.

The Erlenmeyer flask placed in an ice bath after warmed mixture of salicylic acid, acetic anhydride and sulfuric acid had been poured on it.

Crystals of aspirin were slowly precipitated.

The product was then collected by means of suction filtration and it was placed in a paper box.

Tests with the product

Solubility test

To check the solubility of aspirin and salicylic acid in water, ethanol and 10% sodium bicarbonate solution six test tubes were used. In the first three test tubes small crystal of aspirin was added and water in one test tube, ethanol in the other test tube and 10% sodium bicarbonate in the third test tube were added following the addition of aspirin. Aspirin was dissolved only by ethanol but not by water and 10% sodium bicarbonate.

This experiment was repeated to check the solubility of salicylic acid by water, ethanol and 10% sodium bicarbonate in the other three test tubes. The salicylic acid was dissolved by ethanol and 10% sodium bicarbonate but not water

	W a t e r	10% Sodium bicarbonate	E t h a n o l
S a l i c y l i c a c i d	I n s o l u b l e	Soluble, form carbon dioxide bubble	S o l u b l e
A s p i r i n	Insoluble, form suspension	I n s o l u b l e	S o l u b l e

Test for the phenolic hydroxyl group

To each ethanol containing test tubes that aspirin and salicylic acid had been added, one drop of 1% ferric chloride was added. In a test tube which contain ethanol and aspirin the color was not changed, there was no purple as well as violet color formed. In a test tube that contain ethanol and salicylic acid the color became violet.

	1 % f e r r i c c h l o r i d e
A s p i r i n + e t h a n o l	N o p u r p l e , n o v i o l e t c o l o r
S a l i c y l i c a c i d + e t h a n o l	V i o l e t c o l o r

CONCLUSION

In order to prepare aspirin in the laboratory, acetylation of salicylic acid using acetic anhydride in the presence of catalyst is used. The pure aspirin have no a phenol group. Therefore to check the purity of the produced aspirin phenol test is exhibited. During phenol test if a color change to purple or violate is introduced it indicates the existence of phenol in that tested aspirin therefore the pure aspirin is not produced then the experiment should be worked again. If the color change to purple or violate is not introduced it indicates that the produced aspirin is pure i.e., it has no phenol group.